Safety Assessment Report

EU REGULATION (EC) NO. 1223/2009 OF THE 30 NOVEMBER 2009

COSMETIC PRODUCT SAFETY INFORMATION

This document addresses the requirements of the European Union, under the above Regulation, for an assessment of safety of human health for every finished Cosmetic Product before it is placed on the market.

Sponsor: **Faruk Ahmed, FRNHZ LTD, UK.**

Content

PART A

DEFINITION OF THE PRODUCT

PRODUCT NAME:	Freshness Luxury Soft Wet Towels
REFERENCE NO.:	
PCR REFERENCE NO.:	434
MANUFACTURER:	Made in China
PRODUCT TYPE:	Face Cleanser
IFRA CATEGORY:	8

QUANTITATIVE AND QUALITATIVE COMPOSITION OF THE COSMETIC PRODUCT

INCI Name / CI name	INCI / CI Origin	CAS Number	EINECS Number	% w/w
DICHLOROBENZYL ALCOHOL	Synthetic	1777-82-8	217-210-5	0.100000
Function ANTIMICROBIAL, PRESERVATIVE **EU Regulation** V/22				
2-BROMO-2-NITROPROPANE-1,3-DIOL	Synthetic	52-51-7	200-143-0	0.050000
Function PRESERVATIVE **EU Regulation** V/21				
BENZALKONIUM CHLORIDE	Synthetic	63449-41-2 / 68391-01-5 / 68424-85-1 / 85409-22-9	264-151-6 / 269-919-4 / 270-325-2 / 287-089-1	0.080000
Function ANTIMICROBIAL, ANTISTATIC, DEODORANT, PRESERVATIVE, SURFACTANT **EU Regulation** V/54 III/65				
POLYSORBATE 20	Synthetic	9005-64-5		0.250000
Function EMULSIFYING, SURFACTANT **EU Regulation** Not Controlled				
PROPYLENE GLYCOL	Synthetic	57-55-6	200-338-0	0.500000
Function HUMECTANT, SKIN CONDITIONING, SOLVENT, VISCOSITY CONTROLLING **EU Regulation** Not Controlled				

INCI Name / CI name		INCI / CI Origin	CAS Number	EINECS Number	% w/w
CITRIC ACID		Synthetic	77-92-9 / 5949-29-1	201-069-1	0.050000
	Function	BUFFERING, CHELATING, MASKING			
	EU Regulation	Not Controlled			
AQUA		Mineral	7732-18-5	231-791-2	98.770000
	Function	SOLVENT			
	EU Regulation	Not Controlled			
PARFUM		Synthetic			0.200000
	Function	DEODORANT, MASKING, PERFUMING			
	EU Regulation	Not Controlled			

QUALITATIVE FORMULATION (This ingredient list should appear on the label) :

AQUA, PROPYLENE GLYCOL, POLYSORBATE 20, DICHLOROBENZYL ALCOHOL, BENZALKONIUM CHLORIDE, 2-BROMO-2-NITROPROPANE-1,3-DIOL, CITRIC ACID, PARFUM, CITRAL, LINALOOL, LIMONENE.

CALIFORNIA PROPOSITION 65 LIST (Ingredients belonging to this list) :

CITRIC ACID

FRAGRANCE / PARFUME INFORMATION

FRAGRANCE NAME AND CODE: LQ210722 Lemon

MANUFACTURER: Xiamen Amber Daily Chemical Technology Co.

ALLERGENS TO BE DECLARED: Based on the provided IFRA statement by the manufacturer (47th amendment), the product is safe to be used in category 8 (face cleanser) by 100.00, in category 11B (cleansing product in contact with skin):100%

Allergens to be declared (above 0.001% in the finished product): Limonene, Citral, Linalool.

PHYSICAL / CHEMICAL CHARACTERISTICS AND STABILITY OF THE COSMETIC PRODUCT

Product's specification:

Organoleptic properties	
Appearance	Wet towel
Colour	White
Smell	Lemon perfume
Physicochemical criteria	
Specific Gravity	N/A
Viscosity	N/A
pH	@20C: 4.30-4.7

The stability of this product was assessed during 12 weeks under following test's conditions:

4°C Dark, 20°C Dark, 40°C Dark, Window Light & Freeze Thaw Analysis: Colour, Appearance, Odour, pH & Pack Integrity

Colour & appearance:

For samples stored at 4°C, 20°C and freeze-thaw cycling, both the towels and the liquid remained satisfactory for the duration of the trial.
For samples stored at 40°C and window light, both the towels and the liquid remained satisfactory until the 12 week test point when noticeable/significant appearance changes, respectively, were recorded for the towels.

Odour:
Compared to the 4°C standard, samples stored at 20°C, 40°C and freeze-thaw cycling, both the towels and the liquid remained satisfactory for the duration of the trial. Samples stored in the window light remained satisfactory until the 4 week test point when the towel showed a noticeable odour change. By the 8 week test point both the towel and the liquid showed a noticeable odour change.

pH:
Samples stored at 4°C, 20°C and freeze-thaw cycling showed a slight increase in pH. Samples stored at 40°C and window light conditions showed a decrease which was more significant for the window samples towards the end of the trial.

Pack Integrity:

Samples at all storage conditions, except 40°C, remained satisfactory for the duration of the trial. Samples stored at 40°C remained satisfactory until the 12 week test point when the packs appeared noticeably ballooned.

Based on the provided test by ESP(dated 09/05/2018), the product is table for 30 months. There should be a warning as: keep away from heat and direct sunlight.

MICROBIOLOGICAL QUALITY

The efficacy of the preservation of a cosmetic product under development has to be assessed experimentally in order to ensure microbial stability and preservation during storage and use. This is done by challenge testing. The latter is mandatory for all cosmetic products that, under normal conditions of storage and use, may deteriorate or form a risk to infect the consumer.

Based on guidelines published by SCCC 2012, for cosmetics classified in *Category 2*, the total viable count for aerobic mesophyllic microorganisms should not exceed 103cfu/g or 103 cfu/ml of the product. *Pseudomonas aeruginosa*, *Staphylococcus aureus* and *Candida albicans* are considered the main potential pathogens in cosmetic products. These specific potential pathogens must not be detectable in 0.1 g or 0.1 ml of a cosmetic product of *Category 2*. [Based on Colipa1 1997, McEwen et al. 2001, US FDA 2001].

A challenge test consists of an artificial contamination of the finished product, followed
by a subsequent evaluation of the decrease in contamination to levels ensuring the microbial limits established for Categories 1 and 2. The microorganisms used in the challenge test may be issued from official collection strains from any state in the EU to ensure reproducibility of the test and are: *Pseudomonas aeruginosa*, *Staphylococcus aureus* and *Candida albicans.* As mentioned before, the responsible person must guarantee the efficacy of the preservation of his products experimentally by challenge testing. However, as no neither legal nor universal challenge test method is available today, it is up to the responsible person to decide on the details of the test to be used.

Based on the data provided by Melbec Microbiology **(Report number: 5826; 12.06.2018),** the challenge test was based on test method 'Preservative Efficacy Test – ISO 11930:2012'. The result demonstrated that *Aspergilus niger, Candida albicans, Pseudomonas aeruginosa ,Staphyloccus aureus, Escherichia coli* reduced by more than 2 log values from the initial counts at 14 days and there were no increases from 14th day counts at 28th day. There were also no increases in the yeast and mold counts from the initial counts at 14 and 28 days.

The microbiological quality of the product complies with the EU 1223, 2009.

IMPURITIES, TRACES, INFORMATION ABOUT THE PACKAGING MATERIALS

A. Description of the primary packaging and suitability:

The primary packaging is PP. Each towel is packed individually in a plastic bag.

There is no information regarding the product's packaging. However, there is a statement provided by the packaging company (Lin An Jaimeihui Cleaning Co. Ltd) which is stated the packaging materials are notrosamine free.

This statement was approved by the Responsible Person.

B. Note of impurities of the ingredients:

INCI Name / CI name	Notes on Impurities
AQUA	GMP status
PROPYLENE GLYCOL	USP shell (China): 99.9% purity; Lead< 1ppm; Limit of DEG and EG: pass
POLYSORBATE 20	Hangzhou Xianda Antibacterial technology (China): Hydroxy value: 96-106; Residue< 0.25%; Heavy Metal< 0.001; Arsenic< 0.0003%.
PARFUM	IFRA certificate available
DICHLOROBENZYL ALCOHOL	BARQUAT MB 5 0 / PCH-4 2 5(Lonza, China): Quarternary Free amine: 1.25%; Amine HCL: 1.25%
BENZALKONIUM CHLORIDE	BASF (China): Content: 99%; Heavy metals< 0.001%; Arsenic< 0.0003%
2-BROMO-2-NITROPROPANE-1,3-DIOL	Hangzhou Xianda Antibacterial technology (China): purity> 99.0%; Clear solution in water
CITRIC ACID	Hangzhou Xianda Antibacterial technology (China): purity> 95.0%; sulphate ash< 0.05%, chloride< 0.005%; Iron< 0.0005%; Arsenic< 0.0001; Lead< 0.0001%

NORMAL & FORESEEABLE USE

The product is considered as a "Face Cleanser Product". Its purpose is to wash and remove make up. For external application only. This product does NOT intend to treat any medical condition. Oral consumption and inhalation exposure are highly unlikely to be route of exposure for this product.

Specific assessment for cosmetic products intended for use on children under the age of three has not been carried out. Keep the product out of the reach of children.

EXPOSURE TO THE COSMETIC PRODUCTS

Estimated daily exposure levels for different cosmetic product types according to Colipa (Cosmetics Europe) data [SCCNFP/0321/00; Hall et al.2007, 2011].

Adult / Child	Adult
Where Product Used	Face and Neck area
Estimated daily amount applied (g)	1.54
Rinse Off / Leave On	Rinse Off
Retention factor	1
Frequency of use	2/day
Surface area (cm)	565
Calculated relative daily exposure (mg/kg bw/day)	2.5

EXPOSURE TO THE SUBSTANCES

SED = **A** (mg/kg bw/day) x **C** (%)/100 x **DAp** (%)/100

With: SED (mg/kg bw/day) = Systemic Exposure Dosage

A (mg/kg bw/day) = estimated daily exposure to a cosmetic product per kg body weight, based upon the amount applied and the frequency of application: see the calculated relative daily exposure levels for different cosmetic product

C (%) = the Concentration of the substance under study in the finished cosmetic product on the application site

DAp (%) = Dermal Absorption expressed as a percentage of the test dose assumed to be applied in real life conditions

INCI Name / CI name	% w/w	Derm. Abs. %	SED
AQUA	98.770000	100.00	2.47
PROPYLENE GLYCOL	0.500000	100.00	0.0125
POLYSORBATE 20	0.250000	100.00	0.00625
PARFUM	0.200000	100.00	0.005
DICHLOROBENZYL ALCOHOL	0.100000	100.00	0.0025
BENZALKONIUM CHLORIDE	0.080000	100.00	0.002
2-BROMO-2-NITROPROPANE-1,3-DIOL	0.050000	100.00	0.00125
CITRIC ACID	0.050000	100.00	0.00125

UNDESIRABLE EFFECTS AND SERIOUS UNDESIRABLE EFFECTS

Products on the market since: Not provided.

Units of the products sold on the market: Not provided.

Number of undesirable effects: Not provided.

Number of reported serious undesirable effects: Not provided.

This formula was NOT available on the market and No undesirable affects were reported via the manufacturer of the products (Faruk Ahmeh, FRNHZ LTD, UK.) to the safety assessor. Please do note that it is the responsible person's duty to provide information regarding any undesirable/ serious effect to the safety assessor.

No undesirable effects were reported via the manufacturer of the product (Faruk Ahmeh, FRNHZ LTD, UK.) to the safety assessor.

If a serious adverse effect is found, the responsible person SHOULD ask a new product evaluation and notify without delay the information requested in Article 23 of regulation 1223/2009 to the relevant competent authority.

Please note that it is the responsible person's duty to provide information regarding any recorded undesirable effect to the safety assessor.

INFORMATION ON THE COSMETIC PRODUCT

None of the ingredients are: from "animal origin"; contaminated with bovine spongiform encephalopathy (BSE); or classified as carcinogenic, mutagenic or toxic to reproduction (CMR) according to Regulation (EC) No. 1272/2008, cat 1A, 1B, or 2.

The products do NOT contain nanoparticles. The product does not contain any CMR ingredients.

The product is manufactured under GMP condition.

PART B

This assessment is based on information supplied by the client, material manufacturers and from published information in recognised authoritative sources and, whilst best endeavours have been used to check the accuracy of this information, the undersigned cannot be held responsible for any erroneous information supplied to it and used for preparing this assessment.

SAFETY ASSESSMENT CONCLUSION

Under normal or reasonably foreseeable conditions of use, a product made to this formulation is unlikely to produce an abnormally high number of adverse reactions. The product will give consumers the level of safely they can reasonably expect. Recommended warnings should be available on the packaging.

WARNING LABELS AND INSTRUCTION FOR USE

Warning:
Keep away from HEAT and direct sunlight.

In case of irritation, cease application.
For external application only.
Keep away from children.
Do not use on broken or inflamed skin.
Do not use around eye areas.

REASONING

The conclusion is based on two key parameters:

- The Margin of Safety (MoS) of the ingredients based on human exposure to the product and the concentration of each ingredient.
- The toxicological profile of each ingredient(Appendix I, ref for NOEL)

The evaluation of this product has included the calculation of the margin of safety (MoS) of each ingredient. In addition to the margin of safety, other safety factors associated with each ingredient have been taken into consideration, such as: carcinogenicity; photo-toxicity; sensitivity; eye and skin irritation; genotoxicity. Margin of safety was calculated for each ingredient based on 100% dermal absorption. All the calculated margin of safety were well over 100.

The preservatives in this product are checked with related annexes in EU (1223/2009), if there is any limitation of application and all the components & their related concentration are complied with the EU regulations, Annex V.
2-BROMO-2-NITROPROPANE-1,3-DIOL is a secondary amines and and the primary packaging should be nitrosomine free. The assessor relied on the statement provided by the manufacturer (approved by the Responsible Person) that the primary packaging is nitrosamine free.

The product has acceptable stability while being kept away from heat.

The product has acceptable microbiology quality and was manufactured within GMP environment.

The perfume used in this product is considered as safe based on available IFRA certification provided by the Responsible Person. all the allergens above 0.001% should be declared on the label. The product was considered as LEAVE ON

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freshening towel.

The product is a mixture of water (98.8%) and 0.23% preservatives. All the preservatives are allowed to be used in cosmetic products and their relevant concentrations in this product is within the acceptable range. The preservative can be mildly irritant to eyes, hence it is recommended to have a warning as not to be used around eye areas.

Based on the available documents, the product is unlikely to cause considerable adverse effect for consumers, if being used as instructed.

INGREDIENTS	NOEL mg/kg bw/day	MoS	SED (mg/kg bw/day) = Systemic Exposure Dosage
AQUA		0.00000	2.47
PROPYLENE GLYCOL	2,500.00000	200,000.00000	0.0125
POLYSORBATE 20	1,000.00000	160,000.00000	0.00625
PARFUM		0.00000	0.005
DICHLOROBENZYL ALCOHOL	100.00000	40,000.00000	0.0025
BENZALKONIUM CHLORIDE	40.00000	20,000.00000	0.002
2-BROMO-2-NITROPROPANE-1,3-DIOL	31.00000	24,800.00000	0.00125
CITRIC ACID	1,200.00000	960,000.00000	0.00125

Assessor's credentials and approval of part B

Name and address of the safety assessor

I, Dr. Mojgan Moddaresi, am a pharmacist, PhD in cosmetic science, a Chartered Biologist and a Fellow of Royal Society of Biology duly authorised according to **EC Regulation 1223, 2009** to sign the statement. I have taken into consideration the general toxicological profile of each ingredient used, the chemical structure, daily exposure of the product, and margin of safety (MoS)for each ingredient was calculated based on available references on annex I. This product does not pose any harm if it is placed on the market on condition that all the warning labels suggested in part B are implemented.
It is concluded that there are no likely safety hazards from normal use of this product and when used as directed. The product is considered safe for sale in the EU countries.

Date and signature of safety assessor

Dr. MModdaresi

Digitally signed by Dr. MModdaresi
Date: 2018.09.27 13:44:10 +01'00'

Dr. Mojgan Moddaresi PhD, PharmD, FRSB, CBiol, MRSB, MSCS
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Please note that the safety assessment has been based on toxicological data available in the date that safety assessment signed. There is no information regarding consumer adverse reactions based on the data provided by the client to the safety assessor. If the product subsequently causes any significant consumer adverse reactions, the undersigned must be informed and a new evaluation should be made. This certification is for safety assessment of the product. Please note that a responsible person should register the product on the EU portal (European Commission Authentication Service- ECAS).

Appendix I

TOXICOLOGICAL PROFILE OF EACH INGREDIENT

DICHLOROBENZYL ALCOHOL

INCI NAME : DICHLOROBENZYL ALCOHOL

CHEMICAL / IUPAC NAME : 2,4-Dichlorobenzyl alcohol

SUBSTANCE : 2,4-Dichlorobenzyl alcohol

CAS # : 1777-82-8

EC # : 217-210-5

FUNCTION : ANTIMICROBIAL, PRESERVATIVE

PRODUCT TYPE, BODY PARTS :

MAXIMUM CONCENTRATION IN READY FOR USE PREPARATION : 0.15%

ACUTE TOXICITY : LD50 oral in rats 3 g/kg bw, in male mice 2.3 g/kg bw. Subcutaneous LD50 in mice 1.7 g/kg bw

SKIN IRRITATION : On human skin (numbers of subjects not specified) no irritation occurred with 2.5% in a cream formulation on 8 successive days, although a stinging sensation was noticed if damaged skin was treated.
Skin irritation studies carried out in a limited number of guinea pigs showed irritation with 5% and above in polyethyleneglycol, but not with 1%.

OCULAR IRRITATION : In preliminary rabbit studies repeated dermal application of 0.5% caused very slight erythema.

SKIN SENSITISATION : Sensitisation tests in guinea pigs showed no evidence of sensitisation at concentrations of 2% in acetone, 0.1% in water or 2.5 % in a cream formulation.

GENTOTOXICITY : A teratogenicity study in rabbits treated orally with 20 mg/kg/day was negative.

REPEATED DOSE TOXICITY STUDY : A short term (3 wk) oral study in rats with dose levels of up to 500 mg/kg bw and an incomplete, sub-chronic (13 wk) study with 0.36 and 0.72 mg/kg bw were negative.
A short-term (4 wk) oral study in guinea pigs treated with 1.2 mg/kg bw/day was likewise negative.

TOXICOLOKINETIC / DERMAL ABSORPTION : Dermal absorption and excretion in rats was rapid, particularly when formulated in non-polar vehicles. Nearly 90% of the dermal dose applied was excreted in the urine over 96 hours.

REFERENCES : OPINION OF THE SCIENTIFIC COMMITTEE ON COSMETIC PRODUCTS AND NON-FOOD PRODUCTS INTENDED FOR CONSUMERS, SCCNFP/0604/02, final

2-BROMO-2-NITROPROPANE-1,3-DIOL

INCI NAME : 2-BROMO-2-NITROPROPANE-1,3-DIOL
CHEMICAL / IUPAC NAME : 1,3-Propanediol, 2-bromo-2-nitro-
INN NAME : bronopol
SUBSTANCE : Bronopol
CAS # : 52-51-7
EC # : 200-143-0
FUNCTION : PRESERVATIVE

PRODUCT TYPE, BODY PARTS :

MAXIMUM CONCENTRATION IN READY FOR USE PREPARATION : 0.1%

OTHER : Avoid formation of nitrosamines

ACUTE TOXICITY : 455 mg/kg for rats and 590 mg/kg for mice

SKIN IRRITATION : Significant skin and eye irritation was observed in animal studies at 0.5%, but not at 0.1%.

SKIN SENSITISATION : The compound was neither a sensitizer nor a photosensitizer in guinea pig studies.

PHOTOTOXICITY / PHOTOSENSITISATION : The compound was neither a sensitizer nor a photosensitizer in guinea pig studies.

GENTOTOXICITY : This ingredient was neither mutagenic nor teratogenic.

CARCINOGENICITY : This ingredient was neither mutagenic nor teratogenic.

SAFETY COMPARATOR

NOAEL (ORAL, MG/KG/D) : 3-week subchronic dermal toxicity

AUTHORITATIVE SAFETY CONCLUSION : Sensitization was observed in clinical patients at 0.1 and 0.5%, but not in a study on nonclinical volunteers. It is concluded that 5-Bromo-5-Nitro-1,3-Dioxane can be safely used in cosmetic products at concentrations up to and including 0.1 %, except when it may react with amines and amides to form nitrosamines or nitrosamides.

REFERENCES : Final Report on the Safety Assessment of 5-Bromo-5-Nitro-1,3-Dioxane, Volume: 9 issue: 2, page(s): 279-288, International Journal of Toxicology

BENZALKONIUM CHLORIDE

INCI NAME : BENZALKONIUM CHLORIDE

CHEMICAL / IUPAC NAME : Quaternary ammonium compounds, benzyl-C8-18-alkyldimethyl, chlorides

SUBSTANCE : Benzalkonium chloride, bromide and saccharinate (9)

CAS # : 63449-41-2 / 68391-01-5 / 68424-85-1 / 85409-22-9

EC # : 264-151-6 / 269-919-4 / 270-325-2 / 287-089-1

FUNCTION : ANTIMICROBIAL, ANTISTATIC, DEODORANT, PRESERVATIVE, SURFACTANT

PRODUCT TYPE, BODY PARTS :

Rinse-off hair (head) products

MAXIMUM CONCENTRATION IN READY FOR USE PREPARATION : 3% (as benzalkonium chloride)

OTHER : In the final products the concentrations of benzalkonium chloride, bromide and saccharinate with an alkyl chain of C14, or less must not exceed 0.1% (as benzalkonium chloride) For purposes other than inhibiting the development of micro-organismes in the product. This purpose has to be apparent from the presentation of the product

WORDING OF CONDITIONS OF USE AND WARNINGS : Avoid contact with the eyes

SKIN IRRITATION : very slight irritation at the maximum "in use" concentration (0.1 %) and significant irritation when applied at a concentration of 5 %.

OCULAR IRRITATION : slightly irritant

SKIN SENSITISATION : It was not sensitiser to guinea pig or human skin.

GENTOTOXICITY : There was no evidence of mutagenicity according to results obtained in the in vitro Ames tests, chromosome aberration test and sister chromatid exchange assay. There was as well no evidence of carcinogenicity according to the results obtained in a 103 week mice study (upper dose rate: 1.5 mg/kg bw/day).

SAFETY COMPARATOR

NOAEL (ORAL, MG/KG/D) : 28 day repeated oral studies in rats

TOXICOLOKINETIC / DERMAL ABSORPTION : 5%

REFERENCES : OPINION OF THE SCIENTIFIC COMMITTEE ON COSMETIC PRODUCTS AND NON-FOOD PRODUCTS INTENDED FOR CONSUMERS, SCCNFP/0762/03

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POLYSORBATE 20

INCI NAME : POLYSORBATE 20

CAS # : 9005-64-5

FUNCTION : EMULSIFYING, SURFACTANT

DESCRIPTION : Sorbitan, monododecanoate, poly(oxy-1,2-ethanediyl) derivs

ACUTE TOXICITY : Polysorbate 20 is a mixture of laurate esters of sorbitol and sorbitol anhydrides, consisting predominantly of the monoester, condensed with approximately 20 moles of ethylene oxide.
In several acute oral toxicity studies, LD50 of polysorbate-20 was found to be >4600-39000 mg/kg and >25000 mg/kg for rats and mice respectively. Acute dermal LD50 was found to be 3000 mg/kg in guinea pig under occlusive dressing on both intact &abraded skin (CIR 1984).

SKIN IRRITATION : Undiluted polysorbate-20 was found to be non to mild skin irritating when was applied to rabbit skin under occlusive condition (CIR 1984). Further in cumulative irritancy test, undiluted 6% polysorbate-20 was applied for 23 hrs under occlusive conditions and found to be severe-irritant to human skin (CIR 1984).

OCULAR IRRITATION : In three studies, undiluted polysorbate-20 was found to be non to mild irritating when applied to rabbits eyes (CIR 1984).

SKIN SENSITISATION : Formulation containing 6% Polysorbate-20 was found to be non sensitizer, when tested in HRIPT and GPMT for human and guinea pig respectively (CIR 1984).

PHOTOTOXICITY / PHOTOSENSITISATION : In Draize-shelanski test, polysorbate-20 was found to be non-phototoxic/photosensitizer to human skin (FSC, 2007).

GENTOTOXICITY : Polysorbate-20 was found to be non-genotoxic in Ames assay and in vitro mouse lymphoma assay (FSC, 2007).

CARCINOGENICITY : Polysorbate-20 was found to be non carcinogenic in 21 weeks dietary study when administered up to 12500 mg/kg/d (FSC, 2007).
Although there were isolated incidences of benign tumours observed in dermal carcinogenicity studies, however these tumours had the tendency to regress. Based on this, CIR has concluded that polysorbate-20 is non carcinogenic (FSC, 2007).

HUMAN / CLINICAL DATA : NA

REPRODUCTIVE/DEVELOPMENTAL TOXICITY STUDY : In a developmental toxicity study, Polysorbate-20 was administered at 500 and 5000 mg/kg/d concentration to 24-25 pregnant rats on gestation Day 6-15. No difference in the number of corpus lutea, implantations, death rate of pre-implantation embryos and no marked difference in the growth and development of fetuses were observed. Hence under the test condition Polysorbate-20 was concluded to be non-teratogenic.

REPEATED DOSE TOXICITY STUDY : In a chronic toxicity study, diet containing 5% or 10% Polysorbate-20 (approximately 7.5 or 15 g/kg/d, respectively) was administered of to mice for 22 months. Under the test conditions, mild diarrhea was observed in the 10% treated group (FSC, 2007).
In a sub-chronic oral toxicity study, diet containing 1%, 2% and 5% (equivalent to 500, 1,000 and 2,500 mg/kg/d) polysorabate-60 was fed to rats for 90 days and mild diarrhea was observed at 5% concentration (FSC, 2007). Hence based on occurrence of diarrhoea in rats NOAEL was set to be 1000 mg/kg/d for polysorbate-60.
JECFA derived the acceptable daily intake (ADI) of Polysorbates (Polysorbate 20, Polysorbate 60, Polysorbate 65 and Polysorbate 80) group at 10 mg/kg/d, using a safety factor of 100.

Acceptable NOAEL
In line with JECFA determination, NOAEL of 1000 mg/kg/d or alternatively ADI of 10 mg/kg/d could be use for safety calculation of Polysorbate-20.

SAFETY COMPARATOR

NOAEL (ORAL, MG/KG/D) : Pre-clinical

NOAEC (INHALATION, MG/M3 OR PPM) : NA

TOXICOLOKINETIC / DERMAL ABSORPTION : Based on the molecular weight of 1227.72 Da, it is estimated that dermal absorption is expected to be <1%. Hence, as a conservative value of 1% is used for the assessment (R.H. Guy et al 1985).

AUTHORITATIVE SAFETY CONCLUSION : CIR: Reviewed in 1984. Expert panel concluded that polysorbate-20 is safe for use in cosmetic products at present concentrations of use (up to 50%).
JECFA ADI: 0-25 mg/kg (JECFA 1973)
Food safety commission: 10 mg/kg (FSC, 2007)
It is permitted for use as a direct food additive and it is Generally Recognised As Safe (GRAS) (CIR 1984).

REFERENCES : Food Safety Commission (FSC) 2007. Evaluation Report of Food Additives "Polysorbate 20, Polysorbate 60, Polysorbate 65 and Polysorbate 80".
CIR 1984. Final report on the safety assessment of Polysorbates 20, 21, 40, 60, 61, 65, 80, 81 and 85.
JECFA 1973. Polyoxyethylene (20) sorbitan monostearate.
R.H. Guy et al 1985: Percutaneous absorption in man: A kinetic approach. Toxicol. Appl. Pharmacol. 78, 1985.
H. Schäfer, T.E. Redelmeier: Skin barrier – Principles of percutaneous absorption. Karger Verlag 1996.
R. Kroes et al.: Application of the threshold of toxicological concern (TTC) to the safety evaluation of cosmetic ingredients. Food Chem Toxicol. 45/12 2007.
J. Arct et al.: Possibilities for the prediction of an active substance penetration through epidermis. IFSCC magazine 4/3, 2001.

PROPYLENE GLYCOL

INCI NAME : PROPYLENE GLYCOL

CHEMICAL / IUPAC NAME : Propane-1,2-diol

INN NAME : propylene glycol

CAS # : 57-55-6

EC # : 200-338-0

FUNCTION : HUMECTANT, SKIN CONDITIONING, SOLVENT, VISCOSITY CONTROLLING

ACUTE TOXICITY : Propylene glycol (PG) is an aliphatic alcohol as a reaction product of propylene oxide and water.
Acute oral LD50 for propylene glycol was found to be >20100 mg/kg and >18900 mg/kg in rats and mice respectively (CIR 2010 & OECD SIDS 2001). In several studies, acute dermal LD50 was found to be >18000 mg/kg in rabbit (CIR 2010).

SKIN IRRITATION : In a human patch test 100% propylene glycol applied under occlusive open condition, was found to be non-irritating to human (OECD SIDS 2001). In a Draize test 100% propylene glycol was applied on both intact and abraded skin, found to be mild irritating to rabbit skin (CIR 2010).

OCULAR IRRITATION : In nine studies, undiluted propylene glycol was installed in the rabbits eyes and found to non-irritating (OECD SIDS 2001).
A survey of current use concentrations conducted by the Personal Care Products Council (PCPC) reported that propylene glycol has been used at concentrations of 0.0008-99% (highest concentration in products that will be diluted, e.g., bath oils, tablets, or salts) (CIR 2010).

SKIN SENSITISATION : In a human maximization test, 69.5% propylene glycol under occlusive condition was found to be non-sensitizer to human skin (CIR 2010). In a HRIPT, formulation containing 86% propylene glycol, under occlusive condition was found to be non-sensitizer to human skin (CIR 2010).
Propylene glycol generally did not induce sensitization reactions when tested at 12-86% in guinea pigs (CIR 2010). Use studies of deodorants containing 35-73% propylene glycol did not report any potential for eliciting irritation or sensitization (CIR 2010).

PHOTOTOXICITY / PHOTOSENSITISATION : It was found to non-phototoxic (conc. Not specified) in 24 hrs occlusive human exposure (CIR 2010).

GENTOTOXICITY : Propylene glycol was found to be non mutagenic with in vitro chromosomal abrasion (human embryonic cells and human lymphocyte), cell transformation assay (SHE cells), In vivo micronucleus assay (mice, i.p. at 2500, 5000, 10,000, 15,000 mg/kg) and Ames assay (in all strain, ±S9) (CIR 2010 & OECD SIDS 2001).

CARCINOGENICITY : In a long-term toxicity study, propylene glycol was found to non-carcinogenic when diet containing 50,000-ppm propylene glycol was fed to rats for 2 years (CIR 2010 & OECD SIDS 2001).

HUMAN / CLINICAL DATA : NA

REPRODUCTIVE/DEVELOPMENTAL TOXICITY STUDY : In a developmental toxicity study, propylene glycol was administered at 16, 74.3, 345, 1600 mg/kg/d concentration to rats and mice. No developmental toxicity was observed and NOEL for maternal and fetal toxicity was concluded to be highest dose tested that is1600 mg/kg/d fro both rats and mice (CIR 2010).

REPEATED DOSE TOXICITY STUDY : In a sub-acute oral toxicity study, drinking water containing 0.5 to 10% propylene glycol were given to the mice for 14 days, and NOEL was found to be 10,000 mg/kg/d (CIR 2010). In a subchronic 90 days inhalation toxicity study, rats were exposed to 0.16, 1.0, 2.2 mg/L propylene glycol through nose only exposure and NOAEC was found to be 1000 mg/m3 (CIR 2010).
In another long-term study, diet-containing 6250, 12,500, 50,000 ppm propylene glycol were fed to rats for 2 years and NOAEL was found to be 2500 mg/kg/d (OECD SIDS 2001). This NOAEL of 2,500 mg/kg is considered as critical NOAEL for MoS calculation.

Acceptable NOAEL
A NOEL of 2500 mg/kg/d from a 2-year chronic toxicity study in rats can be considered as a safety comparator.

SAFETY COMPARATOR

NOAEL (ORAL, MG/KG/D) : Pre-clinical

NOAEC (INHALATION, MG/M3 OR PPM) : 1000 mg/m3

TOXICOLOKINETIC / DERMAL ABSORPTION : In an in vitro percutaneous absorption study on human abdominal skin, 100% propylene glycol was applied and 23% percutaneous absorption was observed (Fasano et al. 2011). Based on this test result 23% percutaneous absorption may be used in calculating the MoS.

AUTHORITATIVE SAFETY CONCLUSION : CIR: The CIR Expert Panel concluded that propylene glycol is safe as cosmetic ingredients in the present practices of use and concentration and when formulated to be non-irritating (CIR 2010).

JECFA: Propylene glycol is generally recognized as safe (GRAS) as a direct food additive when used in accordance with good manufacturing practices, and it is approved as a direct and indirect food additive.
ADI: 25 mg/kg/day.

REFERENCES : CIR 2010. Final Amended Report of the Cosmetic Ingredient Review Expert Panel. Safety Assessment of Propylene Glycol, Tripropylene Glycol, and PPGs as Used in Cosmetics, June 29, 2010.
OECD SIDS 2001. 1, 2-Dihydroxypropane CAS: 57-55-6. 2001.
Fasano et al. 2011. Dermal penetration of propylene glycols: measured absorption across human abdominal skin in vitro and comparison with a QSAR model; Journal Toxicology in vitro: 2011.

CITRIC ACID

INCI NAME : CITRIC ACID

CHEMICAL / IUPAC NAME : 2-Hydroxy-1,2,3-propanetricarboxylic acid

INN NAME : citric acid

CAS # : 77-92-9 / 5949-29-1

EC # : 201-069-1

FUNCTION : BUFFERING, CHELATING, MASKING

ACUTE TOXICITY : Citric acid has a low acute toxicity by oral application in both rat (LD 50 = 3,000 – 12,000 mg/kg, 3) and mouse (LD 50 = 5,400 mg/kg) (OECD, 2001). The acute dermal LD50 in rabbits was > 5000mg/kg (CIR, 2014)

SKIN IRRITATION : In irritation studies in rabbits, 30% citric acid was not a primary irritant. 60% produced some erythema and edema that subsided with time, and undiluted citric acid produced mild to severe erythema and mild to severe edema (CIR, 2014).

OCULAR IRRITATION : In the Draize test, Citric acid was minimally irritating to rabbit's eyes at a concentration of 10% and mildly irritating at a concentration of 30% (CIR, 2014).

SKIN SENSITISATION : In sensitization testing, a cuticle cream containing 4% citric acid was not an irritant or a sensitizer in humans. 2.5% citric acid produced positive results in skin prick test in 3 of the 91 patients with urticaria or angioedema (CIR, 2014).

GENTOTOXICITY : In several in vitro and in vivo tests citric acid was not mutagenic. The substance was not mutagenic either in bacterial tests with Salmonella typhimurium (Ames test, 2 studies) and Escherichia coli, with and without metabolic activation. A dominant lethal assay with male rats being treated with up to 3 g/kg/d for 5 days was negative (OECD, 2001).

CARCINOGENICITY : In a carcinogenicity study, 20 male rats receiving up to 5% citric acid in the feed (approx. 2 g/kg/d) for 2 years showed no evidence of carcinogenicity. (OECD,2001).

Reproductive/Developmental Toxicity:
In a two-generation 90 days study with male and female rats fed 1.2 % citric acid, no adverse effect on reproductive parameters nor any teratogenicity of dietary citric acid was seen (OECD, 2001).

HUMAN / CLINICAL DATA : See skin sensitization

REPEATED DOSE TOXICITY STUDY : A 2-year chronic oral study in rats being given 5% or 3% citric acid in feed (approx. 2 resp. 1.2 g/kg/d) found slightly decreased growth in the higher dosage group but no tissue abnormalities in the major organs. From the lower dosage a NOAEL of 1200 mg/kg/d was determined (OECD, 2001).

AUTHORITATIVE SAFETY CONCLUSION : CIR Safety Review: Citric Acid, Calcium Citrate, Potassium Citrate, Sodium Citrate and Triethyl Citrate are GRAS for use in food. In addition, Citric Acid is a normal component of the body. Therefore, the CIR Expert Panel focused on the potential for Citric Acid and its salts and esters to cause adverse effects when placed on the skin. At concentrations used in cosmetics and personal care products, Citric Acid and its salts and esters were not eye irritants, nor did they result in skin irritation or sensitisation.

The CIR Expert Panel noted that although Citric Acid could be considered an alpha-hydroxy acid, it is also a beta-hydroxy acid. Structurally, Citric Acid is a tricarboxylic acid. This makes it distinct from the alpha-hydroxy acids previously reviewed by the CIR Expert Panel (for example lactic acid and glycolic acid). The CIR Expert Panel concluded that the concern about increased sun sensitivity resulting from the use of alpha-hydroxy acid containing products was not relevant to products containing Citric Acid and its salts and esters. Based on the available data, the CIR Expert Panel concluded that Citric Acid and its salts and esters were safe for use in cosmetics.
FDA: Citric acid is approved by the FDA as a secondary food additive permitted in food for

human consumption (FDA, 2015).

REFERENCES : CIR Compendium (2013), Abstracts, Discussion and Conclusions of cosmetic ingredient safety assessments. Cosmetic Ingredient Review (CIR), 2013. Washington DC.USA

CIR, (2014).Safety Assessment of Citric Acid, Inorganic Citrates, and Alkyl Citrates Esters as used in cosmetic .Cosmetic Ingredient Review (CIR) 2014, Washington DC, USA.
FDA, (2015) Food for human consumption (21CFR17.280). Part 173. US Food and Drug Administration (2015) Washington DC. USA.

AQUA

INCI NAME : AQUA

CHEMICAL / IUPAC NAME : Water

INN NAME : water

CAS # : 7732-18-5

EC # : 231-791-2

FUNCTION : SOLVENT

ACUTE TOXICITY : As a chemical compound, a water molecule contains one oxygen and two hydrogen atoms that are connected by covalent bonds. Water is a liquid at standard ambient temperature and pressure, but it often co-exists on earth with its solid state, ice; and gaseous state, steam (water vapor).
Water is used in the formulation of most types of cosmetic and personal care products. It can be found in lotions, creams, bath products, cleansing products, deodorants, makeup, moisturisers, oral hygiene products, personal cleanliness products, skin care products, shampoo, hair conditioners, shaving products, and suntan products.

The quality of water used in the production of cosmetics and personal care products, called process water, is monitored according to Good Manufacturing Practices outlined in FDA's Guidance on Cosmetic Manufacturing Practice Guidelines, and in international guidelines on Good Manufacturing Practices known as ISO 22716. Some companies may also comply with the U.S. Pharmacopeia (USP) standards for the purity of water used in drugs, devices and diagnostics published in the Purified Water monograph. USP Purified Water is prepared from water complying with the regulations of the U.S. Environmental Protection Agency (EPA) with respect to drinking water. It contains no intentionally added substances.

Simply water is unlikely to cause irritation, allergy or harm. The source of water should be known, monitored to GMP and either a deionised or high purity grade free from toxins, pollutants and bacteriological contamination should be used in cosmetic products.

PARFUM

INCI NAME : PARFUM

CHEMICAL / IUPAC NAME : Perfume and aromatic compositions and their raw materials

FUNCTION : DEODORANT, MASKING, PERFUMING

Appendix II– Assessor's Address, C.V. and Proof of Credentials

Dr. Mojgan Moddaresi
PhD, PharmD, MSB, FSB, CBiol, MSCS
Personal care Regulatory Ltd., St. John Innovation Centre, Cowley Road, Cambridge, CB4 0WS.

CBiol: chartered Biologist status is recognized as a hallmark of excellence in both th UK and the European Union. Continuing Professional Development (CPD) is the mechanism through which Chartered Biologist can annually demonstrate their credibility and distinction to enable the retention of the title (Ref: https://www.societyofbiology.org/careers-and-cpd/registers/chartered)

FSB: Fellowship of Society of Biology

Professional Body Membership:
Full Member of British Society of Toxicology
Full member of the Society of Cosmetic Scientist
Full member of the Society of Biology,
Member of the Scientific Committee of the Society of Cosmetic Scientists, UK
Member of the Council of the Society of Cosmetic Scientists, UK. (2013-2016)

Education:

2004 - 2010 (PhD in Cosmetics Science)
London College of Fashion, University of the Arts, London (Affiliated with King's College, London, pharmaceutical division, drug delivery group).

Dissertation Title: "The use of nanotechnology in enhancing the efficacy of cosmetics products from natural origins".

1992 - 1998 Doctor of Pharmacy (PharmD).
Faculty of Pharmacy – Tehran, Iran.

Dissertation Title: "Developing topical formulation containing Juniperus Cummonis L. (herbal total extract) and evaluating its efficacy on Acne Vulgarise treatment in clinical study".

2003: ISO 9002 quality management system certification (TUV, Germany).

2005: Cell culture & in vitro models for drug absorption into the skin, Saarbrucken, Germany.

2015: Cosmetic Europe Forum scientific conference, International Scientific and Regulatory Conference, Brussels, Belgium.

Experience:

More than 10 years experience in topical formulations research and development in different industries like cosmetics industry, biopharmaceutical (MedImmune-AstraZeneca, UK), and Pharmaceutical industries.

Posters and Oral Presentation:

2015: 23th IFSCC Conference, Switzerland. "An exploratory review on Cosmetic products' registration: gaps and avenues for global harmonisation".

2015: International Manufacturers and distributors Forum, London. "EU regulation and necessity of due diligence service for the Responsible Person".
http://www.professionalbeauty.co.uk/files/2361_imf_4pp_brochure_2015.pdf

2015: Making Cosmetic, Ricoh arena, Coventry. "Cosmetic-Vigilance: The Collection, Detection, Assessment, Monitoring Undesirable Effects of Cosmetic Products". http://www.making-cosmetics.com/room-a-day-2/

2015: Natural and Organic Products EU, London. "10 important facts to know about the EU regulation". http://www.naturalproducts.co.uk/education/

2015: Future of Formulations in Cosmetics, London. "EU regulation, what to expect in future?" http://www.wplgroup.com/aci/conferences/eu-sff1.asp

2014: International Manufacturers and distributors Forum, London. "Regulation, regulation, Regulation: get it right or it would be costly".

2014: Natural and Organic Products EU, Malmo, Sweden. "Natural products: what to know about the EU compliance procedures? ".http://www.naturalproducts.co.uk/education/

2007: XIIIth COSMODERM – Joint Meeting of ESCAD (and the Hellenic Society of Dermatology and Venereology under the patronage of the European Society for Cosmetic and Aesthetic Dermatology, Athens, Greece. "In vivo assessment of a nanocarrier delivery system with Tocopheryl acetate for topical application".

2007: 8th Skin Forum Annual Meeting, School of Pharmacy, University of London. " A comparison of pig and human skin as an in vitro model for the diffusion of highly lipophilic drugs".

2007:16th International European Academy of Dermatology and Veneorology, Greec. "Development of an effective cosmetically acceptable vitamin E for topical formulation using a novel lipid nanocarrier formulation".

2006: British Pharmaceutical Conference, Manchester. " Vitamin E nanoparticles for dermal applications".

Publication in Trade magazines:

2011, "Peptides in cosmetics, the holy grail or marketing buzzword?" Purehealth, http://viewer.zmags.com/publication/45f03be7#/45f03be7/1.

2010, "Nanocapsulation in cosmetic products, is it the answer to your formulation problem?" http://viewer.zmags.com/publication/bc17d4ab#/bc17d4ab/28.

2009, "Safety of nanoparticles in personal care", Formulation and science technology newsletter, July, issue 7, 3.

2005, "Realising hope in a jar?" SPC magazine (Soap, Perfumery and Cosmetics), 78 (10), 55-57.

Publications in Peer Reviewed Journals:

Moddaresi, M., Brown, M. B., Zhao, Y., Tamburic, S., Jones, S.A., 2010. The role of vehicle–nanoparticle interactions in topical drug delivery. International Journal of Pharmaceutics,400:176-182.

Moddaresi, M., Brown, M. B., Tamburic, S. A., Jones, S. A., 2010. Tocopheryl acetate disposition in porcine and human skin when administered using lipid nanocarriers. Journal of Pharmacy and Pharmacology, 62: 762–769.

Zhao, Y., Moddaresi, M., Jones, S.A., Brown, M. B., 2009. Dynamic hydrofluoroalkane foam to induce nanoparticle modification and drug release in situ, European Journal of Pharmaceutics and biopharmaceutics, 71(3), 522-528.

Moddaresi, M., Tamburic, S., Williams, S., Zhao, Y., Jones, S., Brown, M., The effects of lipid nanocarriers on the performance of topical vehicles in vivo, Journal of Cosmetic Dermatology, Vol 8, Issue 2 (2009)136-143.

Moddaresi, M., S., Tamburic, Brown, M. B., 2006. Vitamin E nanoparticles for dermal applications. Journal of Pharmacy and Pharmacology 58(10), 33.